

Spectral clustering :

↳ It reduces complex multidimensional datasets into clusters of similar data in fewer dimensions.

↳ It uses the connectivity approach to clustering

↳ It uses information from the eigenvalues of special matrices (Affinity matrix, Degree matrix, and Laplacian matrix) derived from graph or data set.

↳ Spectral clustering treats the data clustering as a graph partitioning problem without making any assumption on the form of the data clusters.

↳ It makes ~~no~~ no assumptions about the form of the clusters.

↳ ~~Data~~ The data points in spectral clustering should be connected, but may not necessarily have convex boundary as oppose to the conventional clustering (K-mean, k-medoid...)

↳ It is computationally expensive

↳

Spectral Clustering

→ Spectral clustering is also unsupervised, which is used to find the cluster in a graph.

→ If we are able to find the subgraph of the given graph, then the job of identifying the ~~cluster~~ cluster is done.

→ Why we need to cluster the graph?

↳ Plenty of the data are in the form of graphs.

Ex social networking, Twitter, facebook

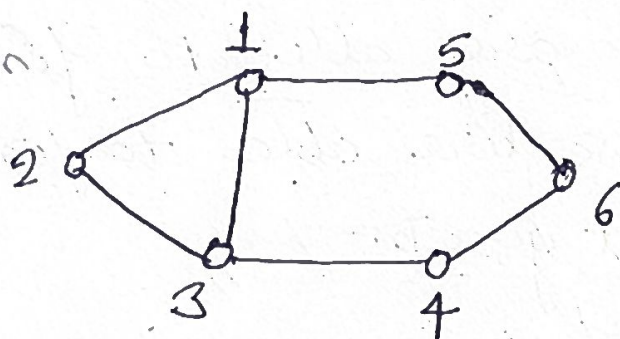
↳ If we are able to find the cluster, then we are able to find the similar interest group.

Spectral clustering has three basic stages:

- 1) Preprocessing: Construct a matrix representation of graph
- 2) Decomposition: Compute eigenvalues and eigen vectors of the matrix
- 3) Grouping: Assign the points to cluster.

Detail step:

- 1) Preprocessing: Build Laplacian matrix L of graph.



First find the ~~Adjace~~ Adjacency matrix

	1	2	3	4	5	6
1	0	1	1	0	1	0
2	1	0	1	0	0	0
3	1	1	0	1	0	0
4	0	0	1	0	0	1
5	1	0	0	0	0	1
6	0	0	0	1	1	0

Find the degree matrix

	1	2	3	4	5	6
1	3	0	0	0	0	0
2	0	2	0	0	0	0
3	0	0	3	0	0	0
4	0	0	0	2	0	0
5	0	0	0	0	3	0
6	0	0	0	0	0	2

Laplacian matrix = $D - A$

	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	2	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

2. Find the eigen values λ and eigen vectors of the Laplacian matrix

- It is proved fact that second smallest eigen values λ_2 partitions the node in such way that it pushes one set of values to one cluster and another set of values to another cluster

	1	2	3	4	5	6
1	0.4	0.3	-1	2.1	0.1	0.2
2	0.2	0.6	0.4	2.6	0.3	0.3
3	0.1	0.3	0.5	0.1	0.2	0.5
4	0.6	-0.3	0.5	0.2	0.4	0.4
5	0.0	-0.3	0.2	0.4	0.5	0.5
6	0	-0.5	0.1	0.2	0.1	0.6

2nd smallest eign vector

Grouping

1 0.3
2 0.6
3 0.3

4 -0.3
5 -0.3
6 -0.3

as an example.